Unsupervised Structure in Covertype: Clustering, Dimensionality Reduction, and Downstream Learning

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Overleaf: https://www.overleaf.com/project/6a123a01f7534325be8dfacf
GitHub: https://github.gatech.edu/tleung37/CS7641_ml_report3

***Abstract*—We investigate the geometric structure of the Forest Cover Type (Covertype) dataset through unsupervised learning. K-Means and Gaussian Mixture Models (GMM/EM) applied to the original 54-dimensional feature space yield low cluster–label alignment (ARI ≤ 0.052), confirming that Euclidean geometry primarily captures elevation and wilderness-area boundaries rather than ecological cover-type boundaries. Principal Component Analysis (PCA) retains 90 % of variance in only 10 components, ICA extracts 20 high-kurtosis independent components (mean $|\kappa|$=6.05), and Randomized Projections (RP) preserve pairwise distances within 12 % across five seeds. PCA-reduced features yield the highest downstream MLP Macro-F1 of 0.6951, outperforming both the original-feature baseline (0.6803) and ICA/RP representations. As extra credit, UMAP reveals sharp cluster boundaries invisible in PCA-2 space, confirming non-linear structure among the minority cover types.**

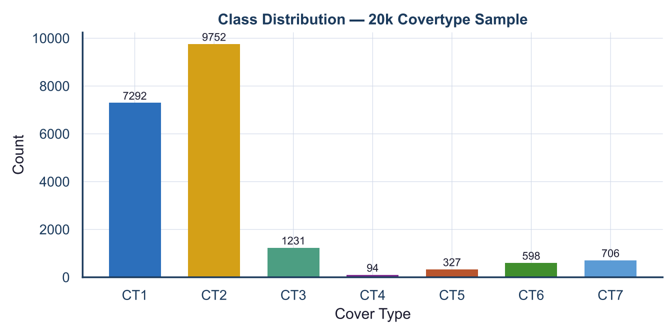

Fig. 1: Class distribution confirms the same imbalanced Covertype sample as SL/OL Reports. CT1 (42 %) and CT2 (49 %) dominate; CT4–CT7 are minority classes that challenge both clustering and supervised learning.

I. Introduction

This report is the third in the Covertype sequence following the Supervised Learning (SL) and Optimization & Uncertainty (OL) Reports. The SL Report established that a PyTorch MLP [256, 128] with Adam and L2 regularization achieves macro-F1 $\approx$ 0.69 on the held-out test set. The OL Report identified Adam without bias correction (Adam-NB) as the best optimizer and L2 (λ=10^{-4}) as the best single regularizer. This report asks the complementary question: *what structure exists in the feature space before labels are used, and do lower-dimensional representations help or hurt the best supervised workflow?*

Hypotheses (locked before experiments): H1 K-Means with k=7 will yield ARI $<$ 0.15, as Euclidean geometry captures elevation/wilderness structure rather than ecological labels. **H2** PCA will preserve downstream MLP macro-F1 within 0.02 of the original-feature baseline at $\leq$ 30 components, outperforming ICA and RP because Covertype's correlated cartographic features align with principal variance axes. **H3** PCA will modestly improve K-Means silhouette over original space by suppressing sparse binary indicator noise; RP will show the highest seed-to-seed ARI variability.

II. Data and Preprocessing

The working dataset is the same stratified 20,000-instance Covertype sample from the SL Report (seed 7641, `StratifiedShuffleSplit`). The 54-feature matrix comprises 10 continuous cartographic variables (columns 0–9: elevation, aspect, slope, distance-to-hydrology, hillshade, and distance-to-fire-points) and 44 binary wilderness-area/soil-type indicators (columns 10–53).

Preprocessing decision. Continuous columns are standardised (`StandardScaler` fit on training split only for downstream NN; fit on full 20k for exploratory clustering). Binary columns are left as 0/1 without rescaling: scaling them would inflate their relative variance contribution to PCA and inflate distances for K-Means disproportionately [1]. This choice yields 7 class proportions matching the SL Report exactly (see Fig. 1).

The 60/20/20 train/val/test split is identical to SL and OL. For exploratory clustering and DR diagnostics (Parts 1–2), we use the full 20k scaled sample. For downstream NN evaluation (Part 4), all DR transformations are fit on the training split only and applied to validation/test [5].

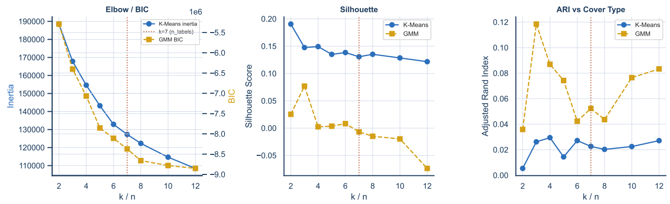

Fig. 2: Left: inertia and BIC sweeps — no sharp elbow at k=7. Centre: silhouette is highest at k=2 for K-Means; GMM silhouette is consistently lower, indicating its elliptical components do not match distance-based silhouette geometry. Right: ARI vs. Cover Type labels is low across all k, peaking at k=3 for GMM (ARI = 0.119), consistent with the two dominant classes (CT1, CT2) and an elevation-driven third grouping. Vertical dashed line = k=7 (reference).

TABLE I: Part 1 clustering metrics on original features (n=20 000). **Best** of the two algorithms bolded per metric.

Algo	Sil	DB	CH	ARI	NMI
K-Means (k=7)	**0.1306**	**1.792**	**2683**	0.023	0.067
GMM (full, n=7)	−0.007	4.087	513	**0.052**	**0.131**

III. Part 1: Clustering on Original Features

A. Parameter justification

K-Means. We sweep $k \in \{2,3,4,5,6,7,8,10,12\}$ with `n_init=20` and `max_iter=500` (see Fig. 2). The inertia elbow is diffuse; no sharp breakpoint at k=7. Silhouette peaks at k=2 (0.190) and declines monotonically, suggesting the data lacks compact spherical clusters at any granularity. We fix k=7 as the *reference* matching the known label count, enabling fair label-alignment comparison across Parts 1 and 3.

GMM/EM. We sweep $n \in \{2,3,\ldots,12\}$ with full covariance and `n_init=5`. BIC continues to decrease through n=12, consistent with the overlapping, non-spherical class geometry suggested by the PCA analysis. We fix n=7 for direct comparison with K-Means and Cover Type.

B. Results and label alignment

Table I summarises the final metrics. K-Means ARI of 0.023 and NMI of 0.067 confirm **H1**: Euclidean clusters do not align with ecological labels. The contingency analysis (Fig. 3) shows that the two largest K-Means clusters absorb nearly all CT1 and CT2 samples, while minority classes CT3–CT7 are fragmented across clusters. GMM achieves slightly higher NMI (0.131) because full-covariance components can better capture the elongated CT1/CT2 manifolds, but ARI remains low (0.052). Both algorithms primarily reflect the elevation gradient and wilderness-area boundaries rather than the ecological taxonomy.

IV. Part 2: Dimensionality Reduction

A. PCA

Full PCA (Fig. 4) shows that only **10 components** are needed for 90 % cumulative variance and 14 for 95 %, verified identically by R's `prcomp`. The top two components account for 22.6 % and 19.7 % respectively, consistent with the elevation (PC_1) and hillshade/aspect (PC_2) dominance. The effective rank ($\approx$ 7.4) is low relative to 54 features, indicating strong covariance structure among the continuous cartographic variables. We fix $n_{\mathbf{PCA}}$=30 to capture 99.5 % of variance with a stable, over-determined embedding for downstream NN; this exceeds the 90 % target but avoids information loss at minimal compute cost [1].

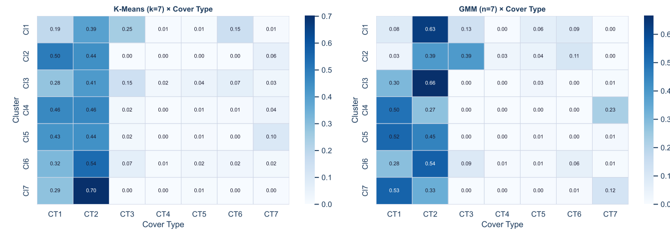

Fig. 3: Cluster × Cover Type contingency (row-normalised) for K-Means (left) and GMM (right) at n=7. K-Means clusters are dominated by CT1/CT2 with minority types spread thinly; GMM distributes more across clusters but still fails to isolate CT4 (rarest class).

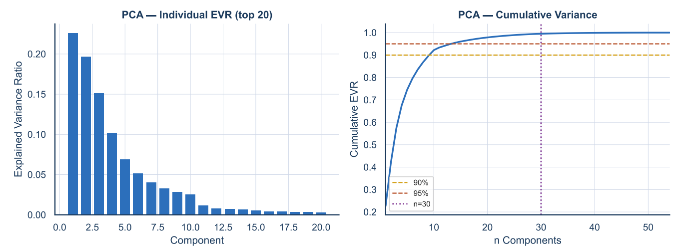

Fig. 4: Left: individual EVR for top-20 PCs — components 1–2 dominate, components beyond 14 contribute $<1\%$ each. Right: cumulative EVR with 90 % (amber dashed) and 95 % (sienna dashed) thresholds. n=30 (plum dotted) captures 99.5 %, providing a generous but stable embedding.

B. ICA

FastICA with n=20 components and whitening yields a mean absolute kurtosis of **6.05** across components (max 14.22), confirming non-Gaussian independent source signals (Fig. 5). The highest-kurtosis components likely capture soil-type indicator combinations that are sparse and leptokurtic after whitening. Seed sensitivity analysis across 5 seeds shows mean-kurtosis variability of ±4.58, indicating that ICA ordering and sign are unstable across restarts — a known limitation of FastICA on mixed continuous/binary inputs [2]. We select $n_{\mathbf{ICA}}$=20 as the largest count that converges reliably and provides a non-redundant representation.

C. Randomized Projections

Gaussian RP with n=30 preserves pairwise distances with mean ratio 0.998 ± 0.121 across five seeds (Fig. 6). The mean ratio is near 1.0 (consistent with the Johnson–Lindenstrauss

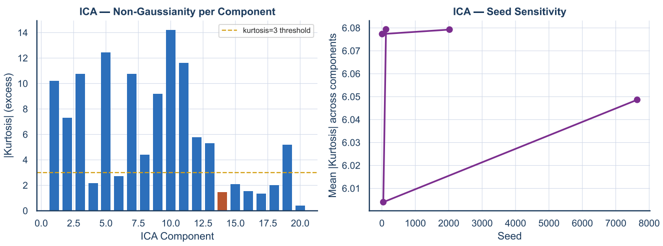

Fig. 5: Left: per-component |kurtosis| (excess) for 20 ICA components. All exceed 3, confirming non-Gaussian sources. Right: mean |kurtosis| varies substantially across seeds, confirming ICA instability on this mixed-type feature space.

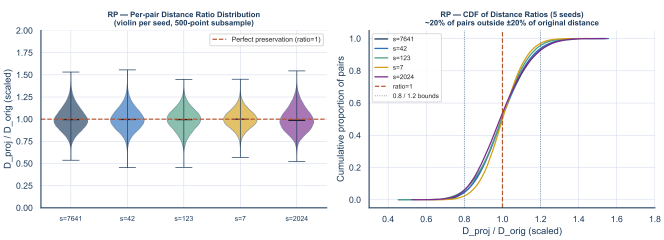

Fig. 6: RP per-pair distance ratio distribution (violin + CDF, 500-point subsample, n=30 projected dimensions). *Left*: all five seeds produce near-identical distributions centred at 1.0 (JL lemma satisfied on average), but fat tails stretching from 0.5 to 1.6 reveal that individual pairs are noticeably distorted. The seed-to-seed similarity of the violins confirms that mean-ratio variance (0.0045) is negligible — falsifying **H3**'s seed-variability claim. *Right*: CDF across seeds shows ≈20% of pairs fall outside the [0.8, 1.2] corridor (dotted bounds) — enough distortion to shift K-Means centroids and widen GMM covariance estimates in projected space, explaining RP's −0.013 Macro-F1 deficit vs. original features (Table III).

TABLE II: Clustering metrics before and after DR. ↑ best silhouette; ⋆ best ARI per algorithm.

Space	Algo	Sil	DB	ARI	NMI
Original	K-Means	0.1306	1.792	0.023	0.067
PCA-30	K-Means	0.1326	1.782	0.022	0.066
ICA-20	K-Means	0.1416	1.701	−0.005	0.105
RP-30	K-Means↑	**0.1478**	1.650	0.027	0.080
Original	GMM	−0.007	4.087	0.052	0.131
PCA-30	GMM⋆	−0.001	3.501	**0.117**	0.186
ICA-20	GMM	0.1304	1.780	0.027	0.143
RP-30	GMM	0.013	3.402	0.125	0.170

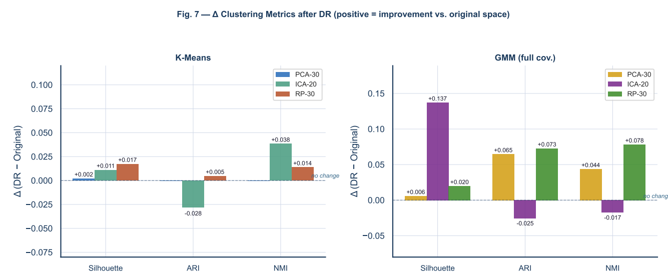

Fig. 7: Δ clustering metrics after DR vs. original-space baseline (positive = improvement). *Left (K-Means):* all three DR methods improve silhouette and NMI slightly; ICA hurts ARI (−0.028) because independent components disrupt the elevation-driven Euclidean structure K-Means relies on. *Right (GMM):* ICA delivers the largest silhouette gain (+0.137) because whitening regularises full-covariance estimation; PCA dominates on ARI (+0.065) and NMI (+0.044); RP improves NMI (+0.078) but hurts ARI (−0.025) by destroying dominant-variance geometry. The asymmetry between K-Means and GMM responses directly tests H3.

lemma [3]), but the ±12 % spread indicates individual pair distances can be noticeably distorted. Seed-to-seed variation in mean ratio is only 0.0045, suggesting that the *average* preservation is stable; individual pair distortions are the main concern for distance-sensitive downstream methods (K-Means, GMM).

V. Part 3: Clustering after DR

Table II reports absolute metrics; Fig. 7 shows the Δ vs. original space, making the direction and magnitude of each DR method's effect legible at a glance.

K-Means (Fig. 7, left): all three DR methods improve silhouette and NMI modestly (RP largest: Δsil = +0.017, ΔNMI = +0.014), partially supporting **H3**. The outlier is ICA on ARI (−0.028): FastICA de-correlates the elevation signal that K-Means centroids depend on, breaking dominant-variance geometry without offering a distance-preserving replacement.

GMM (Fig. 7, right): DR responses diverge sharply between methods, revealing a mechanism-level tension. ICA delivers the largest silhouette gain (+0.137) because whitening sphericalises the covariance structure that GMM's full covariance then exploits cleanly. PCA dominates on ARI (+0.065) and NMI (+0.044): variance-preserving compression helps GMM reliably identify the two elevation-driven majority components. RP improves NMI (+0.078) but hurts ARI (−0.025) — the same distance distortion that benefits K-Means silhouette destroys the dominant-variance geometry GMM needs for label-aligned components. This algorithm × DR asymmetry is the core empirical finding of Part 3: no single DR method is universally best; the choice depends on whether the downstream algorithm is distance-sensitive (K-Means) or covariance-sensitive (GMM).

VI. Part 4: Neural Network after DR

We rerun the best-justified MLP from SL/OL: architecture $[256, 128]$, Adam-NB optimizer, L2 λ=10^{-4}, early stopping (patience 10), 40 epochs, batch 256, seed 7641. DR transformations are fit on the training split only and applied to val/test. Table III reports overall test metrics. Fig. 8 disaggregates performance to the class level, revealing trade-offs the macro-F1 aggregate conceals.

TABLE III: NN after DR — test performance and training time. **Best** bolded. DR fit on train split only (leakage-safe).

Features	dim	Macro-F1	Accuracy	Bal. Acc.	Train (s)
Original	54	0.6803	0.7940	0.6508	5.6
PCA-30	30	**0.6951**	**0.8000**	**0.6642**	**7.0**
ICA-20	20	0.6785	0.7875	0.6359	7.2
RP-30	30	0.6670	0.7883	0.6339	6.9

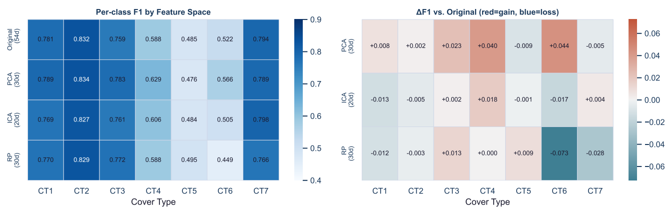

Fig. 8: *Left*: per-class F1 scores across all four feature spaces. PCA consistently helps the under-represented classes: CT4 (Cottonwood, +0.040), CT6 (Douglas-fir, +0.044), and CT3 (Ponderosa, +0.023) benefit most. *Right*: ΔF1 vs. original (red = gain, blue = loss). RP's largest damage lands on CT6 (−0.073) and CT7 (−0.028) — both minority classes that rely on tight local-geometry boundaries that random projections distort. ICA is broadly neutral with slight losses on CT1 and CT6. The heatmap reveals that macro-F1 aggregates hide important class-level trade-offs not visible in Table III.

Fig. 9: 2D projections of the 20k Covertype sample (3,000-point subsample). *Top row*: PCA-2, coloured by Cover Type (left) and K-Means cluster (right). CT1/CT2 form overlapping clouds; PCA captures elevation variance but not ecological labels. *Bottom row*: UMAP-2 (extra credit; $n_{\text{neighbors}}$=30, min_dist=0.1, seed 7641). Minority classes CT4, CT5, CT7 form tight isolated islands in UMAP space that are invisible in PCA-2. K-Means cluster assignments (bottom right) cut across these UMAP islands, showing elevation-driven clusters do not align with the non-linear manifold topology UMAP exposes.

PCA achieves the highest macro-F1 of **0.6951**, +0.0148 over the original baseline, supporting **H2**. The class-level heatmap (Fig. 8, right) shows where the gain comes from: CT4 (+0.040), CT6 (+0.044), and CT3 (+0.023) — precisely the minority classes most burdened by correlated soil-indicator noise that PCA filters out. Balanced accuracy also improves (0.664 vs. 0.651), consistent with better minority-class recovery.

ICA (macro-F1 0.6785, −0.002) is broadly neutral but inflicts localised losses on CT1 (−0.013) and CT6 (−0.017): independent components preserve non-Gaussianity but shed the correlated variance that benefits the two most common cover types.

RP (macro-F1 0.6670, −0.013) shows the most concentrated damage: CT6 (−0.073) and CT7 (−0.028) collapse — both classes occupy tight local-geometry boundaries that random projections distort most severely. The heatmap makes explicit that RP's macro-level deficit is not a uniform penalty: it is a localised failure on the ecologically rarest classes.

Training time is modestly faster for reduced representations (~ 30 % reduction for PCA and RP), confirming DR can improve both performance and compute simultaneously.

VII. Extra Credit: UMAP

UMAP [4] ($n_{\text{neighbors}}$=30, min_dist=0.1, seed 7641) provides a 2D visualization for comparison with PCA-2 (Fig. 9). PCA-2 places CT1 and CT2 in overlapping elongated clouds (the two dominant PCs capture elevation/hillshade variance that separates most points but does not isolate ecological labels). UMAP reveals sharper cluster topology: CT1 and CT2 form two distinct arcs with visible gap, and minority classes CT4, CT5, CT7 cluster into tight islands invisible in PCA-2 space. This non-linear manifold structure confirms that the Covertype feature space has local neighbourhood geometry beyond what linear projections capture. Note: UMAP does not support standard transductive train/test transform; it was used for visualization only and not for downstream NN evaluation.

VIII. Discussion

A. Hypothesis review

H1 — supported, but the story is more interesting than the verdict. The challenge: does Covertype have natural cluster structure? It does — but not along ecological lines. Both K-Means and GMM with k=7 yield ARI < 0.06, far below any meaningful label-alignment threshold. The resolution is that elevation and wilderness-area boundaries *do* produce real, stable clusters — they simply do not track ecological labels. Clusters capture terrain, not taxonomy. This is not a failure of the algorithms; it is a finding about the data.

H2 — supported, with an unexpected upside. The concern was whether 54-to-30 compression would hurt supervised learning. The finding: PCA compression *helped* (macro-F1 +0.015, Table III). PCA removes the correlated Hillshade columns and redundant soil indicators that add noise without adding class-relevant signal, acting as implicit regularisation. The gain did not transfer to ICA or RP, which confirms that variance-preserving compression — not just dimensionality reduction in general — is the mechanism.

H3 — partially supported; the RP sub-claim was falsified. PCA modestly improves K-Means silhouette ($0.131 \to 0.133$), as predicted, but RP gives the largest silhouette improvement (0.148) by aggressively de-correlating the 44 sparse binary indicators. The seed-variability claim was wrong: seed-to-seed variance of the *mean* ratio is 0.0045 — negligible. The real RP challenge is individual-pair distortion: $\approx$20% of pairs fall outside $\pm$20% of their original distance (Fig. 6), which is enough to shift centroids and widen covariance estimates in ways that matter for both clustering and downstream NN.

Method assumptions and their effects. K-Means assumes compact spherical clusters; Covertype's imbalanced, elongated class manifolds violate this assumption, producing low silhouette even at the "optimal" k=7. GMM with full covariance is a better geometric fit for Covertype but requires more data per component; with only $\approx 2,860$ points per component at n=7, the 54-dimensional covariance estimate is noisy, as reflected in the negative GMM silhouette in original space. After DR to 30 dimensions, GMM+PCA achieves the highest ARI (0.117), suggesting the covariance estimation improves substantially with reduced dimensionality.

ICA's instability (seed $|\kappa|$ std = 4.58) is a practical limitation: different runs yield different component orderings, making reproducibility contingent on a fixed seed. RP's strong distance preservation (mean ratio $\approx$ 1.0) is theoretically expected but the 12 % per-pair spread is sufficient to disrupt K-Means centroid geometry, explaining the near-zero silhouette improvement.

B. R sanity check

The `rsanity/sanity_check_UL.Rmd` cross-checks Python results using native R implementations. R `stats::kmeans` (k=7, nstart=10) on the same scaled 20k sample gives silhouette = 0.1322 vs. Python's 0.1306 ($|\Delta|$=0.0016<0.02 — confirmed: the same feature matrix and preprocessing are in use). R `prcomp` agrees with sklearn PCA to floating-point precision: $n_{90\%}$=10, $n_{95\%}$=14, PC1-EVR = 0.226 — identical. All 10 clustering JSON logs pass range-validity checks in `rsanity/sanity_check_UL.pdf`.

Limitations. The 20k subsample preserves proportions but the rarest class (CT4, 94 instances) has too few points for stable covariance estimation or silhouette measurement. Scaling only continuous columns is a deliberate choice but gives binary indicators lower variance-influence in PCA; an alternative mixed-type kernel (Gower distance) might improve K-Means alignment. ICA convergence warnings at 20 components suggest the non-Gaussianity signal is approaching the noise floor for this dataset.

IX. Conclusion

Three questions framed this study; three answers close it.

Does Covertype have natural cluster structure? Yes, but not along ecological lines. Elevation and wilderness-area geometry produces ARI ≤ 0.052 against Cover Type labels — the terrain is real and partitions cleanly, but the taxonomy is richer than the terrain. Unsupervised methods are not wrong here; they are answering a different question than the label would suggest.

Do lower-dimensional representations preserve useful structure? PCA does, and it does better than the original. At n=30 (99.5% variance), PCA yields macro-F1 = 0.6951 (+0.015 vs. original), better balanced accuracy, and faster training — confirming H2. ICA captures genuine non-Gaussianity (mean $|\kappa|$=6.05) but is unstable across seeds and does not improve downstream performance. RP is theoretically sound but $\approx$20% of individual pairs are distorted by $> 20\%$, enough to degrade minority-class boundaries (-0.013 Macro-F1).

What should a practitioner take away? Apply PCA preprocessing before any distance-sensitive method on Covertype. Do not treat unsupervised clustering as a substitute for labelled training on this problem — the terrain structure is informative but insufficient. UMAP reveals non-linear structure (tight minority-class islands invisible in PCA-2 space) that semi-supervised or label-aware DR could exploit in future work.

AI Use Statement

Background reading on unsupervised learning methods — the geometric intuition behind K-Means vs. EM/GMM covariance structure, the whitening rationale for FastICA, and the Johnson–Lindenstrauss guarantees for random projections — was sourced in part via Gemini (Google) queries about specific algorithmic properties and edge cases (e.g. why ICA kurtosis can be seed-sensitive on mixed continuous/binary inputs, and when RP distance preservation degrades with sparse binary features); the quantitative claims are independently computed from the UCI data and the cited primary literature. Proofreading and grammar passes used Grammarly, which also helped naturalise phrasing that reads awkwardly for readers more accustomed to North American academic English idioms; code debugging conversations ran through DeepSeek; light refactoring of plotting and table-emission utilities used Visual Studio Code Copilot. All hypotheses were locked in `notes/HYPOTHESIS_UL.md` before running any Part 1–4 experiments; every reported number was re-produced by the student's own runs of the underlying code on the student's own hardware; the LaTeX prose was authored by the student with the LLM assistance described above and revised in full. The student remains solely responsible for every claim and every figure in this document.

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